

# Product formulas for ballot-polynomial Hankel determinants

Proofs of Cigler's Conjectures 13–15,  
with strict log-concavity and coefficient asymptotics

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19 September 2026

## Abstract

We study the shifted Hankel determinants of the ballot polynomials  $a_m(t) = \sum_{h \leq m/2} \left( \binom{m}{h} - \binom{m}{h-1} \right) t^h$ . An explicit evaluation expresses every normalized determinant as a sum of only  $\lfloor k/2 \rfloor + 1$  monomials, whose coefficients are products of two elementary Vandermonde-type factors. This establishes the sign, support, and positivity asserted in Conjecture 13 of Cigler's 2021 paper. The same products are polynomial in the matrix size on each parity class. Their degree and a negative-index reflection establish the rational generating functions, exact numerator degrees, and reciprocity identities of Conjectures 14 and 15. Moreover, all the residual coefficient sequences are strictly log-concave, and after normalization they converge to a binomial distribution for fixed shift. The proof is self-contained: a block determinant reduces an  $n$ -by- $n$  Hankel determinant to a  $k$ -by- $k$  coefficient determinant; a two-term orthogonal-polynomial connection then collapses its expansion to parity-separated Vandermondes. Exact verification code, symbolic numerator data, and coefficient tables accompany the article.

## 1 The selected problem and the outcome

The problem is a small, sharply delimited part of Johann Cigler's paper *Hankel determinants of middle binomial coefficients and conjectures for some polynomial extensions and modifications* [1]. Section 5, pages 19–21 of version 3, poses three conjectures about the same family of determinants. Their combination is particularly suitable for a direct attack: the moments have a very simple orthogonal-polynomial recurrence, and the predicted support of the normalized determinant is strikingly small.

For integers  $m \geq 0$ , put

$$a_m(t) = \sum_{h=0}^{\lfloor m/2 \rfloor} \left( \binom{m}{h} - \binom{m}{h-1} \right) t^h, \quad \binom{m}{-1} = 0. \quad (1.1)$$

The first polynomials are

$$1, \quad 1, \quad 1+t, \quad 1+2t, \quad 1+3t+2t^2, \quad 1+4t+5t^2.$$

At  $t = 1$ , the sum telescopes to  $a_m(1) = \binom{m}{\lfloor m/2 \rfloor}$ , the middle binomial coefficients, OEIS A001405 [4]. Define

$$\Delta_k(n, t) = \det(a_{k+i+j}(t))_{0 \leq i, j < n}, \quad \delta_k(n, t) = \frac{\Delta_k(n, t)}{t^{\binom{n}{2}}}. \quad (1.2)$$

The empty determinant is 1. We first interpret the quotient over  $\mathbb{Q}(t)$ ; the proof below shows that it is a polynomial, so its value at  $t = 0$  is obtained by polynomial specialization, not division by zero.

Write  $F_k(x, t) = \sum_{n \geq 0} \delta_k(n, t) x^n$ . Here is the exact scope of the three published conjectures, in our indexing. For  $k \geq 1$ , set

$$e_k(n) = \left\lfloor \frac{k(n-1)+1}{2} \right\rfloor, \quad r = \lfloor k/2 \rfloor, \quad d_k = \lfloor k^2/4 \rfloor. \quad (1.3)$$

Conjecture 13 predicts

$$\delta_k(n, t) = (-1)^{e_k(n)} t^{e_k(n)} \rho_k(n, t), \quad \deg_t \rho_k(n, t) = r, \quad (1.4)$$

with positive coefficients. For  $n \geq 1$ , we prove positivity of *every* coefficient of  $t^0, \dots, t^r$ . At the empty-matrix boundary,  $\rho_k(0, t) = t^r$ ; it is best to state that boundary separately rather than include it in a strict-positivity assertion.

Conjecture 14 predicts

$$F_{2r}(x, t) = \frac{A_{2r}(x, t)}{(1 - t^r x)(1 - t^{2r} x^2)^{r^2}}, \quad \deg_x A_{2r} = 2r(r-1) + 1, \quad r \geq 1, \quad (1.5)$$

$$F_{2r+1}(x, t) = \frac{A_{2r+1}(x, t)}{(1 + t^{2r+1} x^2)^{r^2+r+1}}, \quad \deg_x A_{2r+1} = 2r^2 + 1, \quad r \geq 0. \quad (1.6)$$

Conjecture 15 predicts the two reciprocity identities

$$x^{2r(r-1)+1} t^{2r^2(r-1)} A_{2r}(x^{-1}, t^{-1}) = A_{2r}(x, t), \quad (1.7)$$

$$(-1)^r x^{2r^2+1} t^{(2r+1)r^2} A_{2r+1}(x^{-1}, t^{-1}) = A_{2r+1}(x, t). \quad (1.8)$$

We prove all these formulas. The prescribed denominators are not asserted to remain reduced after every specialization of  $t$ .

**Status and attribution.** The cited source is arXiv:2111.14492v3, revised 30 December 2021, where these statements are explicitly labeled conjectures. A targeted literature search conducted for this report, through 19 September 2026, did not locate a subsequent proof of these particular Section 5 statements. That search is not a certification of priority. The mathematical claim here is that the detailed argument below proves the displayed statements; the report has not undergone external peer review or proof-assistant verification. Orthogonal-polynomial reductions of Hankel determinants are established techniques, not a claimed invention of this report; related general results appear in [2, 3]. The useful specialization here is the collapse to one deleted column and the resulting explicit coefficient products.

**Road map.** Section 2 gives the product evaluation. Sections 3–4 prove it from first principles. Section 5 strengthens positivity to strict log-concavity. Sections 6–8 derive the polynomial dependence, asymptotics, generating functions, and reciprocity. The final sections discuss computation, examples, and reproducibility.

## 2 An explicit coefficient product

It is convenient to keep a second weight:

$$a_m(u, t) = \sum_{h=0}^{\lfloor m/2 \rfloor} \left( \binom{m}{h} - \binom{m}{h-1} \right) u^{m-2h} t^h. \quad (2.1)$$

Define  $\Delta_k(n; u, t)$  and  $\delta_k(n; u, t)$  as in (1.2), with these moments. Setting  $u = 1$  recovers the original problem.

**Definition 2.1.** For  $k \geq 1$  and an integer  $n$ , let

$$\mathcal{A}_{k,n} = \begin{cases} \{0, 2, \dots, k\}, & k \text{ even,} \\ \{a : 0 \leq a \leq k, a \equiv n \pmod{2}\}, & k \text{ odd.} \end{cases} \quad (2.2)$$

For  $a \in \mathcal{A}_{k,n}$ , delete one integer from a consecutive interval:

$$S_{k,n,a} = \{n-1, n, \dots, n+k-1\} \setminus \{n+a-1\}.$$

From its even and odd elements form the increasing lists

$$E = (d/2 : d \in S_{k,n,a}, d \text{ even}), \quad O = ((d-1)/2 : d \in S_{k,n,a}, d \text{ odd}).$$

Their lengths are  $e = \lceil k/2 \rceil$  and  $o = \lfloor k/2 \rfloor$ . Define

$$V_E(E) = \frac{\prod_{p < q} (E_q - E_p)(E_q + E_p + 1)}{\prod_{i=0}^{e-1} (2i)!}, \quad (2.3)$$

$$V_O(O) = \frac{\prod_{p=0}^{o-1} (O_p + 1) \prod_{p < q} (O_q - O_p)(O_q + O_p + 2)}{\prod_{i=0}^{o-1} (2i+1)!}, \quad (2.4)$$

$$C_{k,a}(n) = V_E(E)V_O(O). \quad (2.5)$$

All products over empty sets are 1.

**Theorem 2.2** (Product evaluation). *For  $k, n \geq 1$ , every  $C_{k,a}(n)$  is a positive integer, and*

$$\boxed{\delta_k(n; u, t) = (-1)^{k \binom{n}{2}} \sum_{a \in \mathcal{A}_{k,n}} C_{k,a}(n) u^a t^{(kn-a)/2}.} \quad (2.6)$$

*In particular, this normalized determinant contains exactly  $\lfloor k/2 \rfloor + 1$  nonzero monomials before specializing  $u$ .*

To read off  $\rho_k$ , put

$$\varepsilon = \begin{cases} 1, & k \text{ and } n \text{ both odd,} \\ 0, & \text{otherwise.} \end{cases} \quad b_j = C_{k, \varepsilon+2j}(n) \quad (0 \leq j \leq r).$$

Then

$$\rho_k(n, t) = \sum_{j=0}^r b_j t^{r-j}. \quad (2.7)$$

Indeed, the least exponent in (2.6) is  $(kn - \varepsilon - 2r)/2 = e_k(n)$ , and the largest is  $(kn - \varepsilon)/2 = \lfloor kn/2 \rfloor$ . Consequently Theorem 2.2 proves Conjecture 13, including both ends of the support; it proves more than a bound on the degree.

**Example 2.3.** Take  $k = 4$ ,  $n = 2$ . The degree interval is  $1, 2, 3, 4, 5$ , and  $a = 0, 2, 4$  deletes respectively  $1, 3, 5$ . In all three cases  $E = (1, 2)$ , so  $V_E = 2$ . The corresponding odd lists are  $(1, 2), (0, 2), (0, 1)$ , giving  $V_O = 5, 4, 1$ . Thus

$$(b_0, b_1, b_2) = (10, 8, 2), \quad \delta_4(2; u, t) = 10t^4 + 8u^2t^3 + 2u^4t^2.$$

Since  $e_4(2) = 2$ , the residual polynomial is  $\rho_4(2, t) = 2 + 8t + 10t^2$ .

### 3 The moments and their orthogonal polynomials

#### 3.1 A weighted path model and the moment generating function

Consider nonnegative lattice paths that start and end at height 0. An up-step raises the height by 1 and has weight 1; a down-step lowers it by 1 and has weight  $t$ . Horizontal steps are permitted only at height 0 and have weight  $u$ . Length is marked by  $w$ . The unweighted interpretation is the dispersed-Dyck-path model associated with the middle binomial coefficients [1, 4].

Let  $C(v)$  be the Catalan generating function, uniquely determined by  $C(v) = 1 + vC(v)^2$ ,  $C(0) = 1$ . A path is a sequence of either a horizontal step or a positive excursion. A positive excursion contributes  $tw^2C(tw^2)$ . Therefore its generating function is

$$\begin{aligned} M(w; u, t) &= \frac{1}{1 - uw - tw^2C(tw^2)} \\ &= \frac{C(tw^2)}{1 - uwC(tw^2)} = \frac{2}{1 - 2uw + \sqrt{1 - 4tw^2}}. \end{aligned} \quad (3.1)$$

For completeness, the identity  $1 - vC(v) = 1/C(v)$  gives the second expression directly from the first. Expanding it in powers of  $u$  gives

$$M(w; u, t) = \sum_{s \geq 0} u^s w^s C(tw^2)^{s+1}.$$

Lagrange inversion applied to  $C - 1 = vC^2$  gives

$$[v^h]C(v)^{s+1} = \frac{s+1}{2h+s+1} \binom{2h+s+1}{h}.$$

With  $m = s + 2h$ , this is  $\binom{m}{h} - \binom{m}{h-1}$ . Hence  $[w^m]M(w; u, t) = a_m(u, t)$ , proving that the path model has exactly our moments. This also explains the homogeneous weight relation  $u \mapsto \lambda u$ ,  $t \mapsto \lambda^2 t$ .

#### 3.2 A tridiagonal operator

Work initially over the field  $\mathbb{Q}(u, t)$ . On a vector space with basis  $e_0, e_1, \dots$ , define

$$Je_0 = e_1 + ue_0, \quad Je_h = e_{h+1} + te_{h-1} \quad (h \geq 1).$$

For a polynomial  $f$ , let  $L(f) = [e_0]f(J)e_0$ . The path interpretation gives  $L(z^m) = a_m(u, t)$ .

Define monic polynomials

$$U_{-1}(z; t) = 0, \quad U_0(z; t) = 1, \quad U_m(z; t) = zU_{m-1}(z; t) - tU_{m-2}(z; t) \quad (m \geq 1), \quad (3.2)$$

and put

$$P_m(z) = U_m(z; t) - uU_{m-1}(z; t) \quad (m \geq 0). \quad (3.3)$$

Thus  $P_0 = 1$ ,  $P_1 = z - u$ , and  $P_m = zP_{m-1} - tP_{m-2}$  for  $m \geq 2$ . These are rescaled Chebyshev polynomials with a one-term perturbation of the initial condition; no special-function facts are needed below.

An induction gives  $P_m(J)e_0 = e_m$ : the cases  $m = 0, 1$  follow from the definition of  $J$ , and the recurrence advances the identity. Consequently

$$L(z^i P_m) = 0 \quad (i < m), \quad L(z^m P_m) = t^m. \quad (3.4)$$

The second equality follows because the only length- $m$  path from height  $m$  to height 0 consists of  $m$  down-steps. Since the  $P_m$  are monic, (3.4) also gives  $L(P_i P_j) = 0$  for  $i \neq j$  and  $L(P_j^2) = t^j$ . Changing from monomials to the monic  $P_j$ 's in a Gram determinant therefore proves

$$\Delta_0(n; u, t) = \prod_{j=0}^{n-1} t^j = t^{\binom{n}{2}}. \quad (3.5)$$

### 3.3 A self-contained small-determinant reduction

**Lemma 3.1.** *For  $k, n \geq 0$ ,*

$$\delta_k(n; u, t) = (-1)^{nk} \det([z^i]P_{n+j}(z))_{0 \leq i, j < k}. \quad (3.6)$$

*Proof.* Form an  $(n+k)$ -by- $(n+k)$  matrix whose columns are indexed by  $P_0, \dots, P_{n+k-1}$ . Its first  $n$  rows are the functionals  $f \mapsto L(z^i f)$ ,  $0 \leq i < n$ , and its last  $k$  rows are  $f \mapsto [z^i]f$ ,  $0 \leq i < k$ . By (3.4), its upper right block is zero and the upper left block is triangular with diagonal  $1, t, \dots, t^{n-1}$ . Thus its determinant is

$$t^{\binom{n}{2}} \det([z^i]P_{n+j})_{0 \leq i, j < k}.$$

Replace the monic-polynomial columns by the monomial columns  $1, z, \dots, z^{n+k-1}$ . The change-of-basis determinant is 1. The bottom  $k$  rows are now an identity block in the first  $k$  columns and zero elsewhere. Expanding along these rows produces the sign  $(-1)^{nk}$ . The remaining matrix is  $(L(z^{k+i+j}))_{0 \leq i, j < n}$ , whose determinant is  $\Delta_k(n; u, t)$ . Equating the two expressions and using (3.5) proves the assertion.  $\square$

This proof isolates the familiar orthogonal-polynomial reduction rather than treating it as a black box. The remaining work is a special evaluation of its right-hand side.

## 4 Why only a few minors survive

### 4.1 The two-term connection collapses the expansion

Expand the determinant in (3.6) using  $P_{n+j} = U_{n+j} - uU_{n+j-1}$ . A priori there are  $2^k$  choices. Call the first choice in a column *unshifted* and the second *shifted*. If an unshifted column is followed immediately by a shifted column, both selected polynomials are  $U_{n+j}$ , so that term vanishes. Every binary choice pattern other than a shifted prefix followed by an unshifted suffix has such an adjacent pair. Therefore only  $k+1$  terms can survive.

If the shifted prefix has length  $a$ , its scalar coefficient is  $(-u)^a$ , and its increasing list of polynomial indices is precisely  $S_{k,n,a}$ : it is the interval  $n-1, \dots, n+k-1$  with  $n+a-1$  omitted. Each  $U_d$  has the parity of  $d$ . There are  $e = \lceil k/2 \rceil$  even coefficient rows and  $o = \lfloor k/2 \rfloor$  odd rows in (3.6). A term vanishes unless the selected column parities have exactly those multiplicities. Counting the two parities in the consecutive interval gives exactly  $a \in \mathcal{A}_{k,n}$ , leaving  $r+1$  terms.

## 4.2 Vandermonde evaluation of the two blocks

The recurrence (3.2) implies

$$U_d(z; t) = \sum_{h=0}^{\lfloor d/2 \rfloor} (-t)^h \binom{d-h}{h} z^{d-2h} \quad (d \geq 0).$$

In particular,

$$[z^{2i}]U_{2m}(z; t) = (-t)^{m-i} \binom{m+i}{2i}, \quad (4.1)$$

$$[z^{2i+1}]U_{2m+1}(z; t) = (-t)^{m-i} \binom{m+i+1}{2i+1}. \quad (4.2)$$

The binomial factors have useful polynomial forms:

$$\binom{m+i}{2i} = \frac{\prod_{h=0}^{i-1} (m(m+1) - h(h+1))}{(2i)!}, \quad (4.3)$$

$$\binom{m+i+1}{2i+1} = \frac{m+1}{(2i+1)!} \prod_{h=1}^i ((m+1)^2 - h^2). \quad (4.4)$$

For the even block, the row indexed by  $i$ , apart from its factorial factor, is a monic degree- $i$  polynomial in  $m(m+1)$ . Its determinant is therefore the Vandermonde determinant in those quadratic arguments. Since

$$E_q(E_q + 1) - E_p(E_p + 1) = (E_q - E_p)(E_q + E_p + 1),$$

it evaluates to  $V_E(E)$ . For the odd block, first factor  $m+1$  from each column and apply the same argument to the squared arguments  $(m+1)^2$ . This gives  $V_O(O)$ .

This argument also proves integrality without an appeal to divisibility properties of the product expressions:  $V_E$  and  $V_O$  are respectively the determinants of the integer matrices

$$\left( \binom{E_j + i}{2i} \right)_{0 \leq i, j < e}, \quad \left( \binom{O_j + i + 1}{2i + 1} \right)_{0 \leq i, j < o}.$$

When  $n \geq 1$ , all nodes are nonnegative, every node difference is positive, and every sum factor appearing in these products is positive. The factors  $O_j + 1$  are positive as well. Thus both determinants are positive integers.

Factoring the powers of  $(-t)$  in (4.1)–(4.2) contributes the exponent

$$W = \sum E_j + \sum O_j - \binom{e}{2} - \binom{o}{2}.$$

The selected degrees have sum  $kn + k(k-1)/2 - a$ . Since their sum also equals  $2\sum E_j + 2\sum O_j + o$ , substitution gives

$$W = (kn - a)/2. \quad (4.5)$$

### 4.3 The common sign

The positivity calculation must not hide the determinant signs. Let  $R$  be the number of odd-before-even pairs in the row order  $0, 1, \dots, k-1$ , and let  $I(S)$  be the corresponding number in the increasing selected degree list. Permuting both lists into even and odd blocks has sign  $(-1)^{R+I(S)}$ . Including (3.6), the prefix coefficient  $(-u)^a$ , and (4.5), the overall sign exponent is

$$nk + a + (kn - a)/2 + R + I(S). \quad (4.6)$$

When  $a$  is increased by 2 within the admissible set, the selected parity word changes by interchanging one adjacent even-odd pair. Thus  $I(S)$  changes parity. At the same time  $(kn - a)/2$  decreases by 1 and  $a$  increases by 2. The total sign is unchanged. It suffices to check the smallest admissible  $a$ .

For even  $n$ , that index is 0 and both parity words begin with an even element, so  $I(S) = R$ . Formula (4.6) then has parity  $kn/2$ , which agrees with  $k\binom{n}{2}$ . For odd  $n$ , the two cases are

| $k$    | $a_{\min}$ | $R$        | $I(S)$     |
|--------|------------|------------|------------|
| $2r$   | 0          | $r(r-1)/2$ | $r(r+1)/2$ |
| $2r+1$ | 1          | $r(r+1)/2$ | $r(r-1)/2$ |

For  $k = 2r$ ,  $R + I(S) = r^2$  and  $(kn - a)/2 = rn$ , so the resulting sign is positive, as required. For  $k = 2r + 1$ , (4.6) has parity  $(n-1)/2$ , which is the parity of  $k\binom{n}{2}$ . We have therefore proved that every surviving term has the sign  $(-1)^{k\binom{n}{2}}$ . Together with the product evaluation and (4.5), this completes the proof of Theorem 2.2.

## 5 Strict log-concavity: a stronger conclusion

The product evaluation gives more than positivity. The dependence on the deleted node can be made almost one-dimensional.

**Theorem 5.1** (Adjacent ratios). *Let  $k, n \geq 1$ ,  $r = \lfloor k/2 \rfloor$ , and  $b_j = C_{k,\varepsilon+2j}(n)$ . If it is not the case that  $k$  is even and  $n$  is odd, put  $L = n + (n \bmod 2)$ . Then*

$$\frac{b_{j+1}}{b_j} = \frac{(r-j)(L+j)}{(j+1)(L+r+j+1)} \quad (0 \leq j < r). \quad (5.1)$$

*In the remaining case,  $k$  even and  $n$  odd,*

$$\frac{b_{j+1}}{b_j} = \frac{(r-j)(n+j)(n+2j+2)}{(j+1)(n+r+j+1)(n+2j)}. \quad (5.2)$$

*Proof.* All dependence on  $j$  occurs in the parity block from which one node is deleted. In the first set of cases that block is the odd block. Its full node list, before deletion, can be written  $m+1 = L/2 + \ell$ ,  $0 \leq \ell \leq r$ . Deleting  $\ell = j$  removes one column factor and every Vandermonde factor involving that column. Ignoring a positive factor independent of  $j$ , the remaining determinant equals

$$b_j = \frac{K}{j!(r-j)!(L+j)_{r+1}}, \quad K > 0. \quad (5.3)$$

To see the cancellation explicitly, the product of the removed node sums is  $(L + j)_{r+1}/(L + 2j)$ , whereas the removed odd column factor is  $L/2 + j$ . Their quotient supplies the constant factor 2, which is absorbed in  $K$ . Here  $(z)_s = z(z + 1) \cdots (z + s - 1)$ .

For even  $k$  and odd  $n$ , the deleted block is instead the even block, with full nodes  $m = (n - 1)/2 + \ell$ . There is no odd column factor to cancel the omitted diagonal sum. The corresponding expression is

$$b_j = \frac{K(n + 2j)}{j!(r - j)!(n + j)_{r+1}}, \quad K > 0. \quad (5.4)$$

Taking adjacent quotients in (5.3) and (5.4) gives the two stated formulas.  $\square$

**Corollary 5.2** (Strict log-concavity). *For every  $k, n \geq 1$ , the positive coefficient sequence of  $\rho_k(n, t)$  is strictly log-concave at every interior position. Equivalently,*

$$b_j^2 > b_{j-1}b_{j+1} \quad (1 \leq j \leq r - 1). \quad (5.5)$$

*Consequently the coefficient sequence is unimodal. When  $r < 2$ , the strict inequalities have no interior indices and are vacuous.*

*Proof.* For real  $0 \leq y < r$ , define

$$R_L(y) = \frac{(r - y)(L + y)}{(y + 1)(L + r + y + 1)}.$$

Since  $L \geq 1$ ,

$$\frac{d}{dy} \log R_L(y) = -\frac{1}{r - y} + \frac{1}{L + y} - \frac{1}{y + 1} - \frac{1}{L + r + y + 1} < 0.$$

Indeed, the middle two terms sum to a nonpositive number and the other two are negative. The exceptional ratio is  $R_n(y)$  multiplied by  $(n + 2y + 2)/(n + 2y)$ , another positive strictly decreasing function. Thus the positive adjacent ratios  $b_{j+1}/b_j$  strictly decrease in both cases. This is precisely (5.5). Reversing the sequence, as in (2.7), preserves strict log-concavity. A sequence with decreasing positive adjacent ratios increases up to at most one transition and then decreases, giving unimodality.  $\square$

**A useful boundary: not real-rootedness.** The conclusion cannot be upgraded to real-rootedness of every residual polynomial. Example 2.3 gives  $\rho_4(2, t) = 2 + 8t + 10t^2$ , whose discriminant is  $8^2 - 4 \cdot 2 \cdot 10 = -16$ . The coefficient inequalities are therefore not being deduced from real zeros; the adjacent-ratio argument is essential.

**A terminating hypergeometric form.** As an optional compact encoding, in the regular cases

$$\sum_{j=0}^r b_j z^j = b_0 {}_2F_1(-r, L; L + r + 1; -z).$$

This is simply the ratio formula written as a finite sum, so no convergence issue arises. In the exceptional case, take the same expression with  $L = n$  and apply  $1 + (2/n)z d/dz$ . The resulting coefficient multiplier  $(n + 2j)/n$  produces (5.2). The product formula, rather than this notation, remains our primary evaluation.

## 6 Polynomial dependence and coefficient asymptotics

For the rest of the article,  $u = 1$ . The two-weight formula was useful because it made the surviving terms easy to separate; no further specialization is needed to extract the conjectured generating functions.

**Theorem 6.1** (Parity polynomials). *Fix  $k \geq 1$ , a parity of  $n$ , and an admissible deletion index. On that parity class,  $C_{k,a}(n)$  is a polynomial in  $n$  of exact degree  $d_k = \lfloor k^2/4 \rfloor$ . More precisely, with  $a = \varepsilon + 2j$ ,*

$$C_{k,\varepsilon+2j}(n) = K_k \binom{r}{j} n^{d_k} + O_k(n^{d_k-1}), \quad (6.1)$$

where the leading constant is the same on both parity classes, and

$$K_{2r} = \frac{\left(\prod_{\ell=1}^{r-1} \ell!\right)^2}{2^r \prod_{\ell=0}^{2r-1} \ell!}, \quad r \geq 1, \quad (6.2)$$

$$K_{2r+1} = \frac{r! \left(\prod_{\ell=1}^{r-1} \ell!\right)^2}{2^r \prod_{\ell=0}^{2r} \ell!}, \quad r \geq 0. \quad (6.3)$$

For  $r = 0$ , the numerator product in (6.3) is understood as empty, giving  $K_1 = 1$ .

*Proof.* On a fixed parity class, every node is  $n/2$  plus a constant. The node differences in (2.3) and (2.4) are therefore constant, the pairwise sum factors have slope 1, and each odd column factor  $O_j + 1$  has slope  $1/2$ . Thus the degree is

$$\binom{e}{2} + \binom{o}{2} + o = \begin{cases} r^2, & k = 2r, \\ r(r+1), & k = 2r+1, \end{cases}$$

with positive leading coefficient.

The ordinary Vandermonde product for the integer nodes  $0, 1, \dots, r$  with  $j$  deleted is

$$\frac{\prod_{\ell=1}^r \ell!}{j!(r-j)!} = \binom{r}{j} \prod_{\ell=1}^{r-1} \ell!.$$

For even  $k$ , the other parity block has  $r$  consecutive nodes, so the product of all ordinary differences is  $\binom{r}{j} (\prod_{\ell=1}^{r-1} \ell!)^2$ . For odd  $k$ , the other block has  $r+1$  consecutive nodes, contributing one extra factor  $r!$ . In both cases the odd column factors contribute  $2^{-r}$  to the leading coefficient. The factorial denominators combine to  $\prod_{\ell=0}^{k-1} \ell!$ . This gives (6.2)–(6.3), independently of the parity of  $n$ .  $\square$

**Corollary 6.2** (Binomial coefficient profile). *For fixed  $k \geq 1$ ,*

$$\rho_k(n, t) = K_k n^{d_k} (1+t)^r + O_k(n^{d_k-1}) \quad (6.4)$$

coefficientwise as  $n \rightarrow \infty$ . In particular, if  $p_{k,n}(\ell) = [t^\ell] \rho_k(n, t) / \rho_k(n, 1)$ , then

$$p_{k,n}(\ell) \longrightarrow 2^{-r} \binom{r}{\ell} \quad (0 \leq \ell \leq r).$$

For each fixed positive real  $t$ ,

$$\delta_k(n, t) = (-1)^{k \binom{n}{2} t^{e_k(n)}} \left( K_k n^{d_k} (1+t)^r + O_{k,t}(n^{d_k-1}) \right). \quad (6.5)$$

*Proof.* Sum (6.1) in (2.7) and use the binomial theorem. The normalization at  $t = 1$  has leading coefficient  $2^r K_k > 0$ . The determinant asymptotic follows from (1.4).  $\square$

The first leading constants are

$$K_1 = 1, \quad K_2 = \frac{1}{2}, \quad K_3 = \frac{1}{4}, \quad K_4 = \frac{1}{48}, \quad K_5 = \frac{1}{576}.$$

There is also an exact, rather than merely asymptotic, way to obtain all lower-order corrections. Expand the finite products (2.3)–(2.4) on each parity class. The expansion terminates at degree  $d_k$ ; there is no unproved remainder estimate or analytic interchange of infinite expansions. The two parity classes can have different subleading terms even though their leading terms agree.

The boundary  $n = 0$  is compatible with these same parity polynomials. In the minor expansion, every shifted prefix of positive length contains  $U_{-1} = 0$ , while the unshifted term has indices  $0, \dots, k-1$  and determinant 1. Equivalently, at  $n = 0$ ,  $C_{k,0}(0) = 1$  and all other admissible coefficients are zero. This observation is important: the generating function below includes its correct constant term, without a separately added correction.

## 7 Rational generating functions and recurrences

**Theorem 7.1** (The prescribed denominators). *The generating functions  $F_k(x, t)$  have the rational forms (1.5) and (1.6), with numerators  $A_k(x, t) \in \mathbb{Z}[t, x]$ . At this stage the numerator degrees are bounded by the denominator degrees minus 1; their exact smaller degrees will follow in Section 8.*

*Proof.* For  $k = 2r$ , put  $d = r^2$ . The product formula gives

$$t^{-rn} \delta_{2r}(n, t) = \sum_{j=0}^r C_{2r, 2j}(n) t^{-j}.$$

By Theorem 6.1, the right side is a degree- $d$  polynomial on each parity class of  $n$ , with the same leading coefficient on the two classes. It can therefore be written

$$P(n) + (-1)^n R(n), \quad \deg P \leq d, \quad \deg R \leq d-1,$$

where the polynomial coefficients lie in  $\mathbb{Q}(t)$ . For a polynomial  $p(n)$  of degree at most  $s$ , the series  $\sum_{n \geq 0} p(n) y^n$  has denominator dividing  $(1-y)^{s+1}$ . One proof is to expand  $p(n)$  in the basis  $\binom{n}{j}$  and use  $\sum_{n \geq 0} \binom{n}{j} y^n = y^j / (1-y)^{j+1}$ . Thus the denominator of  $F_{2r}$  divides

$$(1 - t^r x)^{d+1} (1 + t^r x)^d = (1 - t^r x) (1 - t^{2r} x^2)^d,$$

which is precisely (1.5).

For  $k = 2r + 1$ , put  $d = r(r + 1)$ . The sign in (2.6) is  $(-1)^m$  at both  $n = 2m$  and  $n = 2m + 1$ . More explicitly,

$$\delta_k(2m, t) = (-t^k)^m \sum_{j=0}^r C_{k,2j}(2m) t^{-j},$$

$$\delta_k(2m + 1, t) = (-t^k)^m \sum_{j=0}^r C_{k,1+2j}(2m + 1) t^{r-j}.$$

Each displayed sum is a polynomial in  $m$  of degree at most  $d$ . Generating the even and odd subsequences with  $y = -t^k x^2$  gives a denominator dividing  $(1 + t^k x^2)^{d+1}$ , proving (1.6).

The resulting rational functions are proper in  $x$ , as is immediate from the binomial-basis expressions. Finally, the prescribed denominator has constant term 1 and lies in  $\mathbb{Z}[t, x]$ . Multiplying it by the original series in  $\mathbb{Z}[t][[x]]$  gives coefficients in  $\mathbb{Z}[t]$ . Rationality says that all but finitely many coefficients vanish, so  $A_k \in \mathbb{Z}[t, x]$ , not merely  $\mathbb{Q}(t)[x]$ .  $\square$

**Corollary 7.2** (Constant-coefficient recurrences). *Let  $\mathbf{E}f(n) = f(n + 1)$ . For all  $n \geq 0$ ,*

$$(\mathbf{E} - t^r)^{r^2+1}(\mathbf{E} + t^r)^{r^2} \delta_{2r}(n, t) = 0, \quad (7.1)$$

$$(\mathbf{E}^2 + t^{2r+1})^{r^2+r+1} \delta_{2r+1}(n, t) = 0. \quad (7.2)$$

*Proof.* The factor  $\mathbf{E} - \lambda$  lowers the polynomial degree of a sequence  $\lambda^n p(n)$  by one. Apply this observation to the even case in the preceding proof. In the odd case, apply the same argument to each parity subsequence with  $\mathbf{E}^2$  in place of  $\mathbf{E}$ .  $\square$

These are valid annihilating recurrences, without a claim of minimality. For example,  $\delta_1(n, t) + t\delta_1(n - 2, t) = 0$  for  $n \geq 2$ , while the even-shift order is  $2r^2 + 1$ , independent of the matrix size  $n$ .

## 8 Negative indices, exact degrees, and reciprocity

Rationality alone does not explain the numerator degrees in Conjecture 14, or the reversal symmetries in Conjecture 15. Both come from a reflection of the same deleted-index products. The negative indices used here are an algebraic continuation of the coefficient sequence, not matrices of negative size.

### 8.1 A canonical bilateral continuation

Extend  $U_d$  to every integer index by the recurrence (3.2), working with invertible  $t$ . It immediately yields

$$U_{-d-2}(z; t) = -t^{-d-1} U_d(z; t) \quad (d \geq 0), \quad U_{-1} = 0. \quad (8.1)$$

The coefficient identities (4.1)–(4.2) continue to hold for every integer  $m$ , if binomial coefficients with negative upper argument are interpreted polynomially:  $\binom{z}{s} = z(z - 1) \cdots (z - s + 1)/s!$ . One may check this directly from (8.1); the two quadratic factorizations (4.3)–(4.4) are polynomial identities and remain valid as well.

Now use the right side of (3.6), with  $P_d = U_d - U_{d-1}$ , to define  $\delta_k(n, t)$  for all integer  $n$ . The prefix expansion, parity count, Vandermonde evaluation, and sign calculation continue verbatim. Thus (2.6), with  $u = 1$  and the product (2.5), remains valid at all integer  $n$ , although its coefficients

need not be positive when  $n < 1$ . On each parity class the product is exactly the polynomial already identified in Theorem 6.1. Consequently this is the exponential-polynomial continuation determined by the nonnegative values; it is not a new choice of values made to force a symmetry.

**Lemma 8.1** (A negative-index gap). *For every  $k \geq 1$ ,*

$$\delta_k(-1, t) = \delta_k(-2, t) = \cdots = \delta_k(-(k-1), t) = 0. \quad (8.2)$$

*For  $k = 1$ , the displayed list is empty.*

*Proof.* Let  $n = -h$ ,  $1 \leq h \leq k-1$ . Before a deletion, the degree interval in the prefix expansion contains  $-2, -1, 0$ . If a degree other than  $-1$  is deleted, the zero column  $U_{-1}$  remains. If  $-1$  is deleted, the proportional columns  $U_{-2} = -t^{-1}U_0$  and  $U_0$  both remain. Every term vanishes.  $\square$

## 8.2 Reflection of the product coefficients

**Lemma 8.2** (Coefficient reflection). *For an admissible index and any integer  $m$ ,*

$$C_{k,a}(-m) = (-1)^{d_k} C_{k,k-a}(m-k). \quad (8.3)$$

*Proof.* The map  $d \mapsto -d - 2$  sends the selected degree set  $S_{k,-m,a}$  to  $S_{k,m-k,k-a}$ , with the order reversed. It sends an even node  $E$  to  $-E - 1$ , and an odd node  $O$  to  $-O - 2$ . Each even pairwise sum factor changes sign, as does each odd pairwise sum factor and each odd column factor. Reversing the order makes the ordinary difference factors agree with their counterparts. The total sign is therefore

$$(-1)^{\binom{e}{2} + \binom{o}{2} + o} = (-1)^{d_k}.$$

The factorial denominators are unchanged, proving the identity.  $\square$

**Theorem 8.3** (Bilateral determinant reflection). *Put  $s_k = (-1)^{\lceil k/2 \rceil}$  and  $B_k = \binom{k+1}{2}$ . Then for every integer  $m$ ,*

$$\boxed{\delta_k(-m, t) = s_k t^{-B_k} \delta_k(m-k, t^{-1}).} \quad (8.4)$$

*In particular,*

$$\delta_k(-k, t) = s_k t^{-B_k} \neq 0. \quad (8.5)$$

*Proof.* Apply Lemma 8.2 termwise in (2.6) and replace  $a$  by  $k-a$ . This bijects the two admissible index sets. The exponents of  $t$  match because

$$-B_k - \frac{k(m-k) - (k-a)}{2} = \frac{-km - a}{2}.$$

The ratio of the two determinant signs has exponent, modulo 2,

$$d_k + \frac{k\{m(m+1) - (m-k)(m-k-1)\}}{2} \equiv d_k - \frac{k^2(k+1)}{2} \equiv \left\lceil \frac{k}{2} \right\rceil.$$

For the last congruence, substitute  $k = 2r$  or  $k = 2r + 1$ . This gives (8.4). Setting  $m = k$  uses  $\delta_k(0, t^{-1}) = 1$  and gives (8.5).  $\square$

### 8.3 Reading a rational function at infinity

We need an elementary link between this continuation and rational functions. It is stated explicitly to justify the exact degree argument.

**Lemma 8.4** (Expansion at infinity). *Suppose  $f_n$  is a finite sum of terms  $p(n)\lambda^n$ , with  $p$  a polynomial and  $\lambda \neq 0$ , defined for every integer  $n$  over a field of characteristic zero. Then the rational function  $F(x) = \sum_{n \geq 0} f_n x^n$  has Laurent expansion*

$$F(x) = - \sum_{m \geq 1} f_{-m} x^{-m} \quad (8.6)$$

at infinity.

*Proof.* For  $f_n = \lambda^n$ , expand

$$\frac{1}{1 - \lambda x} = - \frac{(\lambda x)^{-1}}{1 - (\lambda x)^{-1}} = - \sum_{m \geq 1} \lambda^{-m} x^{-m}.$$

Apply  $(x d/dx)^j$  to obtain the assertion for  $f_n = n^j \lambda^n$ , and use linearity.  $\square$

For our sequences the lemma applies over an algebraic extension of  $\mathbb{Q}(t)$  if necessary. In the even case the bases are  $t^r$  and  $-t^r$ . In the odd case one can use the two square roots of  $-t^{2r+1}$ . All final identities lie in  $\mathbb{Q}(t)(x)$ , so this intermediate field extension introduces no restriction.

**Theorem 8.5** (Exact numerator degrees). *The numerators in Theorem 7.1 have exactly the degrees asserted in Conjecture 14:*

$$\deg_x A_{2r} = 2r(r-1) + 1, \quad \deg_x A_{2r+1} = 2r^2 + 1.$$

*Proof.* By Lemma 8.4, the negative-index gap and its first nonzero successor give

$$F_k(x, t) = -s_k t^{-B_k} x^{-k} + O(x^{-k-1}) \quad (x \rightarrow \infty), \quad (8.7)$$

as a formal Laurent expansion. Thus, writing  $F_k = A_k/Q_k$  with the prescribed denominator, we have  $\deg_x A_k = \deg_x Q_k - k$ . The two denominator degrees are  $2r^2 + 1$  and  $2(r^2 + r + 1)$ , giving the required formulas. No assertion that  $A_k$  and  $Q_k$  are relatively prime is needed: the degree difference describes the rational function at infinity even when both polynomials contain a common factor.  $\square$

### 8.4 The reciprocal identities

**Theorem 8.6** (Generating-function and numerator reciprocity). *As rational functions,*

$$F_k(x, t) = -s_k t^{-B_k} x^{-k} F_k(x^{-1}, t^{-1}). \quad (8.8)$$

Consequently  $A_k$  satisfies (1.7) and (1.8), proving Conjecture 15.

*Proof.* Insert (8.4) in (8.6):

$$F_k(x, t) = -s_k t^{-B_k} \sum_{m \geq 1} \delta_k(m - k, t^{-1}) x^{-m}.$$

For  $1 \leq m < k$ , every summand vanishes by the negative-index gap. Reindex the remaining terms by  $n = m - k$  to obtain (8.8). Equality of the Laurent expansions implies equality of the rational functions.

For even  $k = 2r$ , direct factorization of the denominator gives

$$Q_{2r}(x^{-1}, t^{-1}) = (-1)^{r^2+1} t^{-r-2r^3} x^{-1-2r^2} Q_{2r}(x, t). \quad (8.9)$$

Combine this with (8.8) and  $B_{2r} = 2r^2 + r$ . The signs cancel, and

$$A_{2r}(x^{-1}, t^{-1}) = t^{-2r^2(r-1)} x^{-2r(r-1)-1} A_{2r}(x, t).$$

This is (1.7). For odd  $k = 2r + 1$ , put  $d = r(r + 1)$ . Now

$$Q_k(x^{-1}, t^{-1}) = t^{-k(d+1)} x^{-2(d+1)} Q_k(x, t). \quad (8.10)$$

Since  $B_k = k(r + 1)$ , substitution gives

$$A_k(x^{-1}, t^{-1}) = (-1)^r t^{-kr^2} x^{-2r^2-1} A_k(x, t),$$

which is (1.8).  $\square$

**Corollary 8.7** (Leading numerator coefficients). *All numerators have constant term  $A_k(0, t) = 1$ . Moreover,*

$$\begin{aligned} [x^{2r(r-1)+1}] A_{2r}(x, t) &= t^{2r^2(r-1)}, \\ [x^{2r^2+1}] A_{2r+1}(x, t) &= (-1)^r t^{(2r+1)r^2}. \end{aligned}$$

Their degrees in  $t$  are respectively  $2r^2(r - 1)$  and  $(2r + 1)r^2$ .

*Proof.* The constant term comes from  $Q_k(0, t) = F_k(0, t) = 1$ . Compare the opposite extreme coefficients in the reciprocity identity to obtain the leading terms. Since  $A_k$  is a polynomial in  $t$ , reciprocity also says that no  $t$ -exponent can exceed the stated reciprocity exponent. The leading terms attain that bound.  $\square$

All uses of  $t^{-1}$  above take place over  $\mathbb{Q}(t)$ . The resulting numerator identities are polynomial identities, so they extend to  $t = 0$  after their displayed monomial factors are multiplied out. The rational description of  $F_k$ , whose denominator has constant term 1 in  $x$ , is likewise valid as a formal generating function under every parameter specialization.

## 9 Examples and computational consequences

### 9.1 Small numerator polynomials

Finite convolution of the product formula with the denominator produces

$$\begin{aligned} A_1(x, t) &= 1 + x, \\ A_2(x, t) &= 1 + x, \\ A_3(x, t) &= 1 + (1 + 2t)x - t^2(2 + t)x^2 - t^3x^3, \\ A_4(x, t) &= 1 + (1 + 3t + t^2)x + t^2(1 + t)(1 + 4t)x^2 \\ &\quad + t^4(1 + t)(4 + t)x^3 + t^6(1 + 3t + t^2)x^4 + t^8x^5. \end{aligned}$$

These agree with the small examples in the original paper [1, Section 5]. The signs of the numerator coefficients are not uniformly positive:  $A_3$  already has negative coefficients. The positivity theorem concerns  $\rho_k$ , not  $A_k$ . The first few degree data are

| $k$          | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8   |
|--------------|---|---|---|---|----|----|----|-----|
| $d_k$        | 0 | 1 | 2 | 4 | 6  | 9  | 12 | 16  |
| $\deg_x Q_k$ | 2 | 3 | 6 | 9 | 14 | 19 | 26 | 33  |
| $\deg_x A_k$ | 1 | 1 | 3 | 5 | 9  | 13 | 19 | 25. |

The accompanying `data/numerators.json` contains all numerator polynomials through  $k = 10$ , with integer coefficient triples  $(i, j, c)$  representing  $cx^i t^j$ . It also contains the prescribed denominators in the same format, making the data independent of a particular computer-algebra syntax.

## 9.2 Residual polynomials

For example, the product formula gives

$$\begin{aligned}
 \rho_2(5, t) &= 3 + 3t, \\
 \rho_3(6, t) &= 12 + 16t, \\
 \rho_4(7, t) &= 120 + 270t + 150t^2, \\
 \rho_5(8, t) &= 1650 + 4400t + 3025t^2, \\
 \rho_6(9, t) &= 55055 + 195195t + 231231t^2 + 91091t^3.
 \end{aligned}$$

The coefficients can be large even when the polynomial degree is small. This is precisely the situation in which an explicit product is more useful than expansion of a large determinant.

## 9.3 An exact coefficient algorithm

The evaluation has two natural implementations. The most transparent one independently evaluates (2.5) for each of the  $r + 1$  terms. A faster one evaluates  $b_0$  once and then uses the rational adjacent ratios of Theorem 5.1. Every resulting update is an integer, so exact division is an internal consistency check.

In arithmetic-operation counts, the first coefficient uses  $O(k^2)$  pairwise factors and the remaining coefficients use  $O(k)$  updates. The determinant dimension  $n$  no longer determines a matrix-elimination cost. This is an arithmetic count, not a unit-cost assertion for arbitrarily large integers: coefficient bit lengths grow with  $n$  and  $k$ , and a numerical value of  $t^{e_k(n)}$  may itself require many bits. The reference implementation provides both coefficient algorithms and checks them against each other.

For generating functions, write  $Q_k(x, t) = \sum_{j=0}^D q_j(t)x^j$  and let  $N = D - k$ . Then

$$A_k(x, t) = \sum_{n=0}^N \left( \sum_{j=0}^{\min(n, D)} q_j(t) \delta_k(n - j, t) \right) x^n. \quad (9.1)$$

Thus  $N + 1$  explicitly known determinant values suffice to construct the entire numerator. The proof establishes that every later coefficient of  $Q_k F_k$  vanishes; a finite computation is not being used to guess that vanishing.

**Why this particular attack succeeds.** The decisive structural feature is not a large computation. It is the short connection  $P_m = U_m - uU_{m-1}$ . Selecting terms in adjacent columns produces duplicate columns except for a single switch point. The problem is reduced to choosing which one of  $k + 1$  consecutive indices is omitted. Parity then cuts the count roughly in half, and the Chebyshev coefficient arrays have quadratic Vandermonde evaluations. Positivity, parity-polynomial growth, and reflection are three aspects of that same product, rather than unrelated lucky patterns.

## 10 Exact verification and reproducibility

The proof above is the basis for the general statements. The accompanying computations serve a different purpose: they check indexing, signs, normalization, boundary values, examples, and the implementation through independent routes. No floating-point comparison is used in the tests.

The standard-library suite completed **32,666 exact checks**. Its principal groups are summarized in Table 1; the full machine-readable record is `data/verification.json`.

| Check family                                                | Number |
|-------------------------------------------------------------|--------|
| Independent shifted Hankel determinants: grid               | 1,092  |
| Independent shifted Hankel determinants: seeded cases       | 200    |
| Independent unshifted Hankel determinants                   | 91     |
| Positive individual coefficients                            | 10,080 |
| Adjacent-ratio identities                                   | 8,640  |
| Strict log-concavity inequalities                           | 7,260  |
| Agreement of direct and ratio-based coefficient algorithms  | 1,440  |
| Parity-polynomial degree and leading-term checks            | 70     |
| Negative-index reflection identities                        | 2,352  |
| Negative-index gap values                                   | 360    |
| Vanishing generating-function tail coefficients             | 984    |
| Numerator reciprocity and exact degree, rational parameters | 96     |
| Non-real-rooted counterexample                              | 1      |
| Total                                                       | 32,666 |

Table 1: Exact checks performed by `code/verify.py`. The count records individual assertions, not independent theorems.

The determinant grid uses  $1 \leq k \leq 12$ ,  $0 \leq n \leq 12$ , and seven integer pairs  $(u, t)$ , including zero and negative weights. The seeded additional cases have  $1 \leq k \leq 20$  and  $0 \leq n \leq 18$ . Each direct determinant is evaluated from the ballot-sum moments by fraction-free Bareiss elimination with pivoting; it does not use the small orthogonal-polynomial determinant or the product formula internally. At  $t = 0$ , the comparison is made with  $t^{(n)}\delta_k(n; u, t)$ , never by dividing a numerical determinant by zero.

The coefficient inequalities and ratios cover  $1 \leq k \leq 24$ ,  $1 \leq n \leq 60$ . Finite differences check both parity-polynomial degrees and their exact leading terms through  $k = 10$ . The bilateral and generating-function checks use  $1 \leq k \leq 16$  and  $t \in \{1, 2, -1/2\}$ . Reflection is checked for  $0 \leq m \leq 48$ ; generating-function coefficients are checked through  $x^{\deg Q_k + 12}$ .

A second suite adds **24 independent bivariate symbolic determinant identities** for  $1 \leq k \leq 6$ ,  $1 \leq n \leq 4$ , retaining both  $u$  and  $t$  as indeterminates. It also checks 135 symbolic tail coefficients and the complete symbolic reciprocity identity for each  $1 \leq k \leq 10$ . The symbolic determinant calculations use SymPy; the latter generating-function calculations use sparse integer

polynomial arithmetic and retain symbolic  $t$ . The saved report identifies the actual runtime and SymPy versions.

## 10.1 Running and rebuilding

From the extracted archive directory, run

```
python code/verify.py
python -m pip install -r requirements.txt
python code/verify_symbolic.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first test needs only Python 3.9 or later. The second test requires SymPy; the tested version is recorded in `requirements.txt`. The  $\text{\LaTeX}$  source uses ordinary packages distributed with TeX Live or MiKTeX. All references are included in the source, so there is no BibTeX or external bibliography-generation step. Shell and PowerShell build scripts are included as conveniences.

The principal artifacts are `article.tex`, `article.pdf`, `code/hankel.py`, the two verification scripts, and the data files. `data/rho_coefficients.csv` supplies coefficients for  $1 \leq k \leq 12$ ,  $1 \leq n \leq 20$ . No downloaded paper or font file is needed to build or test the package.

## 11 Conclusion and scope

The explicit deleted-index product resolves the three stated mathematical claims as follows. Positivity and the precise support prove Conjecture 13. Parity-polynomial dependence gives the predicted denominators, while the negative-index gap supplies the exact numerator degrees, completing Conjecture 14. Reflection of the even and odd nodes yields the rational functional equation and hence Conjecture 15. The adjacent-ratio formulas further prove strict log-concavity, and the leading products give a binomial coefficient profile with explicit constants.

The scope is exactly these three source conjectures and the additional results stated here. No conclusion is claimed about the other conjectures in the source paper, universal real-rootedness, or a reduced denominator after every parameter specialization. Historical priority remains subject to the qualification in Section 1; the proof and the verification artifacts are provided for independent examination.

## References

- [1] Johann Cigler. *Hankel determinants of middle binomial coefficients and conjectures for some polynomial extensions and modifications*. arXiv:2111.14492v3, revised 30 December 2021. In particular, Section 5, pp. 19–21, Conjectures 13–15. <https://arxiv.org/abs/2111.14492v3>.
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